THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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ANSWER KEY & MARKING SCHEME SET 1

Chapter 10: **The Other Side of Zero** | Class 6 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** -5	Same signs: add and keep sign; different signs: subtract.
2	**(a)** -7	Add the opposite.
3	**(b)** -6	Right is positive, left is negative.
4	**(d)** 8	Count the integers strictly between.
5	**(a)** -9	Add the rise.
6	**(d)** 0	$n + (-n) = 0$.
7	**(d)** 50	Debt is negative.
8	**(c)** -17	Same signs: add and keep sign; different signs: subtract.
9	**(c)** 5	Add the opposite.
10	**(c)** 9	Right is positive, left is negative.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	5 yellow tokens and 3 red tokens (1); Cancel pairs: the value is 2 (1).	1+1
12	-4 is to the right of -7 on the number line (1); $-4 > -7$ (1).	1+1
13	$7 - 3 = 4$; $3 - 7 = -4$ (1); The two answers are opposites of each other (1).	1+1
14	6 red tokens and 4 yellow tokens (1); Cancel pairs: the value is -2 (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	First two: -4 (1); Then add 5 (1); Result 1 (1).	1+1+1
16	At noon: $-4 + 9 = 5$°C (1); In the evening: $5 - 8 = -3$°C (1); Below 0°C (1).	1+1+1
17	$5 - 9 = -4$ (1); $-4 + (-3) = -7$ (1); Check using the number line (1).	1+1+1
	OR: First two: -10 (1); Then add 15 (1); Result 5 (1).	1+1+1
18	At noon: $-6 + 8 = 2$°C (1); In the evening: $2 - 5 = -3$°C (1); Below 0°C (1).	1+1+1
	OR: $-4 - 6 = -10$ (1); $-10 + 8 = -2$ (1); Check using the number line (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	After rising: $-80 + 30 = -50$ m (1); After diving: $-50 - 25 = -75$ m (1); Total distance travelled $= 30 + 25 = 55$ m (1); Below sea level (2).	1+1+1+2
	OR: Total $= 120 + (-80) + 50 + (-30) = 60$ (1); Best day: Mon (₹120) (1); Worst day: Tue (₹-80) (1); Positive total means overall profit (2).	1+1+1+2
20	After rising: $-120 + 45 = -75$ m (1); After diving: $-75 - 60 = -135$ m (1); Total distance travelled $= 45 + 60 = 105$ m (1); Below sea level (2).	1+1+1+2
	OR: Total $= -60 + 90 + (-40) + 70 = 60$ (1); Best day: Tue (₹90) (1); Worst day: Mon (₹-60) (1); Positive total means overall profit (2).	1+1+1+2

Mistakes to avoid:

- Saying $-7 > -3$ because 7 is greater than 3; on the number line -7 lies to the left of -3.
- Writing $-4 - (-6) = -10$; the correct value is $-4 + 6 = 2$.
- Dropping the sign of the bigger number when adding integers with different signs, e.g. $(-9) + 5 = 4$ instead of -4.
- Taking the successor of a negative integer as one smaller (successor of -1 is 0, not -2).
- Finding a fall or rise as initial − final in the wrong order; use final − initial and read the sign.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET 2

Chapter 10: **The Other Side of Zero** | Class 6 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 9	Count the integers strictly between.
2	**(a)** -14	Add the rise.
3	**(a)** positive	Distance is positive.
4	**(d)** -25	Debt is negative.
5	**(d)** -7	Same signs: add and keep sign; different signs: subtract.
6	**(c)** 13	Add the opposite.
7	**(a)** -14	Right is positive, left is negative.
8	**(c)** 11	Count the integers strictly between.
9	**(a)** -12	Add the rise.
10	**(a)** does not exist	There is no smallest negative integer.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	-3 is to the left of 2 on the number line (1); $-3 < 2$ (1).	1+1
12	$-5 - 2 = -7$; $2 - (-5) = 7$ (1); The two answers are opposites of each other (1).	1+1
13	2 red tokens and 5 red tokens (1); Cancel pairs: the value is -7 (1).	1+1
14	-9 is equal to -9 on the number line (1); $-9 = -9$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	First two: 7 (1); Then add (-9) (1); Result -2 (1).	1+1+1
16	At noon: $-2 + 7 = 5$°C (1); In the evening: $5 - 10 = -5$°C (1); Below 0°C (1).	1+1+1
17	$7 - (-3) = 10$ (1); $10 + (-12) = -2$ (1); Check using the number line (1).	1+1+1
	OR: First two: 2 (1); Then add (-5) (1); Result -3 (1).	1+1+1
18	At noon: $-9 + 12 = 3$°C (1); In the evening: $3 - 7 = -4$°C (1); Below 0°C (1).	1+1+1
	OR: $-10 - (-4) = -6$ (1); $-6 + 9 = 3$ (1); Check using the number line (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	After rising: $-60 + 20 = -40$ m (1); After diving: $-40 - 15 = -55$ m (1); Total distance travelled $= 20 + 15 = 35$ m (1); Below sea level (2).	1+1+1+2
	OR: Total $= 200 + (-150) + (-50) + 80 = 80$ (1); Best day: Mon (₹200) (1); Worst day: Tue (₹-150) (1); Positive total means overall profit (2).	1+1+1+2
20	After rising: $-90 + 50 = -40$ m (1); After diving: $-40 - 30 = -70$ m (1); Total distance travelled $= 50 + 30 = 80$ m (1); Below sea level (2).	1+1+1+2
	OR: Total $= -90 + (-20) + 130 + 40 = 60$ (1); Best day: Wed (₹130) (1); Worst day: Mon (₹-90) (1); Positive total means overall profit (2).	1+1+1+2

Mistakes to avoid:

- Saying $-7 > -3$ because 7 is greater than 3; on the number line -7 lies to the left of -3.
- Writing $-4 - (-6) = -10$; the correct value is $-4 + 6 = 2$.
- Dropping the sign of the bigger number when adding integers with different signs, e.g. $(-9) + 5 = 4$ instead of -4.
- Taking the successor of a negative integer as one smaller (successor of -1 is 0, not -2).
- Finding a fall or rise as initial − final in the wrong order; use final − initial and read the sign.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 10: **The Other Side of Zero** | Class 6 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 80	Debt is negative.
2	**(a)** 0	Same signs: add and keep sign; different signs: subtract.
3	**(a)** -19	Add the opposite.
4	**(d)** 11	Right is positive, left is negative.
5	**(c)** 7	Count the integers strictly between.
6	**(d)** 0	Add the rise.
7	**(c)** -1	Closest to 0 on the left.
8	**(a)** 0	Debt is negative.
9	**(c)** -16	Same signs: add and keep sign; different signs: subtract.
10	**(a)** 13	Add the opposite.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$-8-(-3)=-5$; $-3-(-8)=5$ (1); The two answers are opposites of each other (1).	1+1
12	7 yellow tokens and 7 red tokens (1); Cancel pairs: the value is 0 (1).	1+1
13	5 is to the right of -12 on the number line (1); $5>-12$ (1).	1+1
14	$4-(-6)=10$; $-6-4=-10$ (1); The two answers are opposites of each other (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	First two: -8 (1); Then add 4 (1); Result -4 (1).	1+1+1
16	At noon: $-5+5=0$°C (1); In the evening: $0-6=-6$°C (1); Below 0°C (1).	1+1+1
17	$3-11=-8$ (1); $-8+(-2)=-10$ (1); Check using the number line (1).	1+1+1
	OR: First two: -11 (1); Then add 14 (1); Result 3 (1).	1+1+1
18	At noon: $-8+15=7$°C (1); In the evening: $7-9=-2$°C (1); Below 0°C (1).	1+1+1
	OR: $-6-(-7)=1$ (1); $1+(-5)=-4$ (1); Check using the number line (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	After rising: $-150+70=-80$ m (1); After diving: $-80-40=-120$ m (1); Total distance travelled $=70+40=110$ m (1); Below sea level (2).	1+1+1+2
	OR: Total $=75+(-125)+100+(-25)=25$ (1); Best day: Wed (₹100) (1); Worst day: Tue (₹-125) (1); Positive total means overall profit (2).	1+1+1+2
20	After rising: $-45+15=-30$ m (1); After diving: $-30-30=-60$ m (1); Total distance travelled $=15+30=45$ m (1); Below sea level (2).	1+1+1+2
	OR: Total $=-45+60+(-80)+95=30$ (1); Best day: Thu (₹95) (1); Worst day: Wed (₹-80) (1); Positive total means overall profit (2).	1+1+1+2

Mistakes to avoid:

- Saying $-7>-3$ because 7 is greater than 3; on the number line -7 lies to the left of -3.
- Writing $-4-(-6)=-10$; the correct value is $-4+6=2$.
- Dropping the sign of the bigger number when adding integers with different signs, e.g. $(-9)+5=4$ instead of -4.
- Taking the successor of a negative integer as one smaller (successor of -1 is 0, not -2).
- Finding a fall or rise as initial − final in the wrong order; use final − initial and read the sign.

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